DAKSH PANCHAL

+91 9910153794 | dakshpanchal08@gmail.com | LinkedIn | GitHub

Education

Delhi Technological University **Aug 2024 – May 2028**
B.Tech Computer Science and Engineering | GPA: 8.87/10 (up to 3rd sem) *Delhi*
JEE Main 2024: AIR 11,214 (99.3 Percentile) **2024**
Goodley Public School: CBSE XII (95.4 Percentage) **2024**

Personal Projects

Mars Exploration Drone Software Stack | **GitHub** **2026**

- Designed a Pixhawk–Raspberry Pi flight software stack with MAVLink-based dual-pilot control (<100ms latency), telemetry routing at 50Hz, and failsafe manual override; stress-tested over 50+ flight hours with no critical failures
- Implemented an onboard processing pipeline combining 64MP imaging with a 3-sensor atmospheric suite (SHT45, BME688, particulate sensor) and dual-RF telemetry for redundant, fault-tolerant communication

MineSync – Distributed Minecraft Server Orchestration | **GitHub** **2026**

- Built a self-managed Minecraft 1.21.11 hosting system for 4 players with on-demand host switching, GitHub-based world synchronization, and lock-based coordination to prevent concurrent server sessions
- Developed cross-platform Bash/Batch automation for server lifecycle management, NetBird private-network connectivity without port forwarding, and automatic save-and-sync workflows for reliable multiplayer access across variable networks

Chess Engine – Game AI System | **GitHub** **2025**

- Engineered a C++ chess engine with alpha-beta pruned minimax search (depth 8, <200ms per move) leveraging efficient data structures (bitboards) for move generation across the full game tree
- Integrated a neural network evaluator trained on 500K+ positions, improving move quality ∼18% over handcrafted heuristics; shipped a cross-platform Flutter UI with modular design for real-time engine interaction

Experience – Research Team: Unmanned Ground Vehicle

Mars Exploration Drone | **ISDC 2026** | **Repository** | **Report** **2026**

- Led MAVLink-based flight software integration on Pixhawk–Raspberry Pi hardware, achieving <100ms control latency and 50Hz telemetry; zero safety-critical failures across the full 50+ hour ISDC 2026 test campaign
- Integrated 64MP imaging with heterogeneous sensor fusion across temperature, gas, and particulate channels (SHT45, BME688); dual-RF telemetry achieved zero communication dropout across all pre-competition test flights

Autonomous Ground Vehicle | **IGVC 2025 USA** | **Repository** | **Report** **2025**

- Implemented a ROS2-based autonomy stack with SLAM and DWA local planning, integrating multi-sensor fusion (LiDAR, stereo camera, IMU) to achieve 95% waypoint accuracy in dynamic outdoor environments

Achievements

Top 10 Global Rank – ISDC 2026 | International Space Drone Challenge **Manipal, India | 2026**
Top 10 Global Rank – IGVC 2025 | Autonomous Navigation Challenge **Michigan, USA | 2025**

Position of Responsibility

UGV-DTU Research Team **Aug 2024 – Present**

- **Software Head (2025–Present):** Lead a team of 15+ software engineers; own software architecture, codebase standards, technical direction, and Agile sprint planning across all competition deliverables
- **Software Engineer (2024–2025):** Developed core autonomy, navigation, and sensor fusion modules; maintained Git-based version control workflows and code review processes across the team

Technical Skills

Languages: C++17/20, Python, Go, JavaScript/TypeScript
Frameworks & Libraries: FastAPI, Next.js, React, Node.js, OpenCV, TensorFlow
Databases: MongoDB, PostgreSQL
Systems & DevOps: Linux, Docker, Git, CI/CD, REST APIs, Microservices, System Design
Embedded & Robotics: ROS2, MAVLink, SLAM, Kalman/EKF, Gazebo, Arduino, UART/SPI/I2C